Lecture-7
Fluid Kinematics II:
Decomposition of fluid motion

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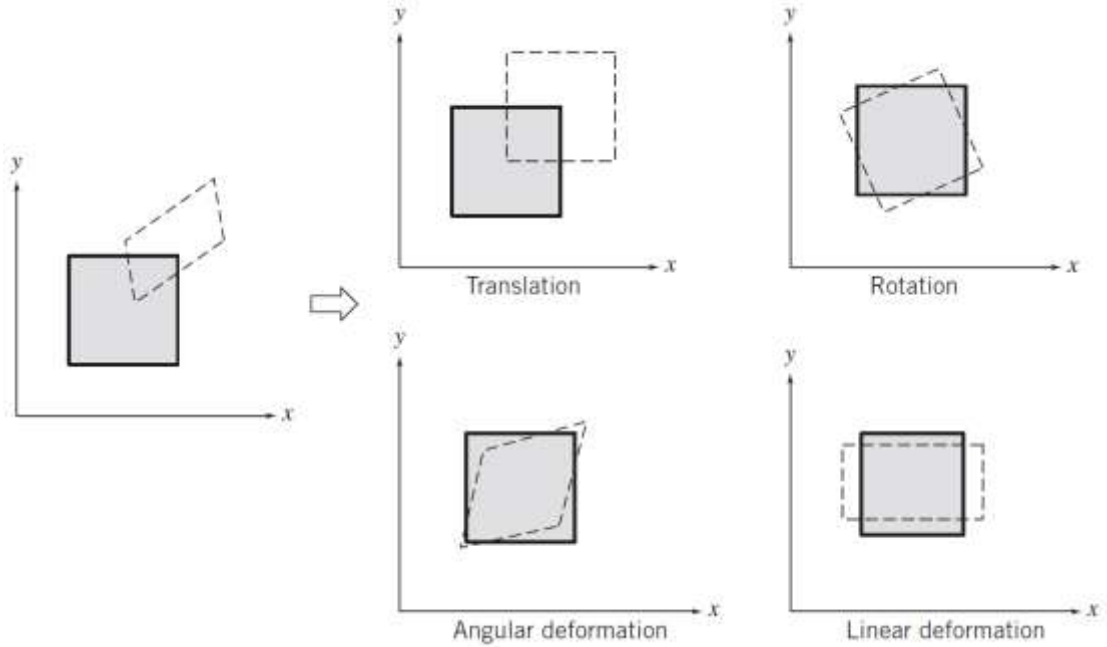
Learning Objectives

By the end of this lecture, you should be able to

1. Deconstruct a complex fluid velocity field into its **four primary kinematic components**.
2. Partition the velocity gradient tensor into its **symmetric (deformation) and anti-symmetric (rotation) components**.
3. Calculate **volumetric dilation** and **shear strain rates** given a 2D or 3D velocity field.
4. Understand the concept of a **free vortex**.

Motion of a fluid particle

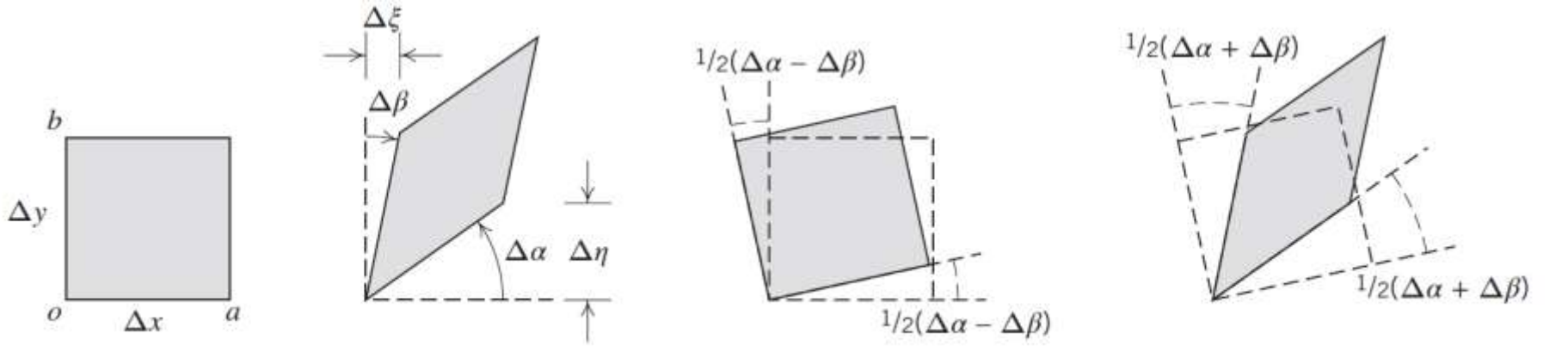
- The motion of a fluid particle can be decomposed into 4 components:
- Translation, Rotation
- Linear Deformation – Sides of the particle **stretch/contract**
- Angular Deformation – The **initial angle** between the sides of the particle **changes**



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Rotation and angular deformation

- The rotation and deformation of a fluid element are described using the **velocity gradient**.



(*a*) Original particle (*b*) Particle after time Δt (*c*) Rotational component (*d*) Angular deformation component

$$\vec{\omega} = \frac{1}{2}\left[\hat{i}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right) + \hat{j}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) + \hat{k}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\right]$$

$$\text{Volume dilation rate} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \nabla \cdot \vec{V}$$

Revisiting the learning objectives

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